

3 Yr. Degree/4 Yr. Honours 6th Semester Examination, 2026 (CCFUP)

Subject : Chemistry

Course : CHEM 6012 (MAJOR)

(Organic Chemistry)

Time: 3 Hours

Full Marks: 60

*The figures in the margin indicate full marks.
Candidates are required to give their answers in their own words
as far as practicable.*

1. Answer any ten questions from the following:

2×10=20

- (a) CO₂ is an IR inactive molecule.—Justify or Criticise.
- (b) Give an example of green organic reaction having 100% atom economy.
- (c) Mention two advantages of using SC-CO₂ (supercritical CO₂) as solvent for organic synthesis.
- (d) Write down the most stable conformation of trans-1, 3-dimethyl cyclohexane. Is it optically active?
- (e) Give the product when furan reacts with diazomethane.
- (f) Why is TMS used as the reference compound in ¹H-NMR spectroscopy?
- (g) Explain the terms bathochromic shift and hypsochromic shift in UV-spectroscopy.
- (h) Mention two characteristics of water that make it a benign solvent.
- (i) What happens on the absorption of infrared and UV radiation by a molecule?
- (j) Ethylene and acetylene do not show C-C multiple bond stretching frequency in IR. —Why?
- (k) What happens when piperidine is subjected to Hoffmann exhaustive methylation? Show the steps.
- (l) Identify the NMR active and inactive nuclei from the following : ²H, ¹⁴C, ¹⁹F, ³¹P.
- (m) Convert : Quinoline to 8-hydroxyquinoline.
- (n) What is 'Fermi resonance'?
- (o) *cis*-4-hydroxycyclohexanecarboxylic acid lactonises on heating but the *trans*-isomer does not.—Explain.

2. Answer any four questions from the following:

- (a) (i) Predict the structure of the molecule from the following spectral data.

Molecular formula : $C_4H_{10}O$.

I. R. (cm^{-1}): 3350 (disappear in presence of D_2O), 2960.

1H -N.M.R.: δ 0.89 (6H, d), 1.3 (1H, m), 3.31 (2H, d), 4.33 (1H, s) ppm

U.V.: No λ_{max} above 220 nm.

- (ii) Explain why 1H -NMR spectrum of CH_3OH in CCl_4 shows two singlets but in $(CD_3)_2SO$ it shows a doublet and a quartet. 3+2

- (b) (i) The position of λ_{max} of acetone shifts in different solvents:

279 nm in hexane, 272 nm in ethanol and 264.5 nm in water—explain why?

- (ii) The chemical shift values of protons of a compound is irrespective of the spectrophotometer frequency—justify the statement with suitable example. 3+2

- (c) (i) Briefly explain why Fluorous Biphasic Solvent (FBS) is ideal for hydroformylation of olefins (mention two points only).

- (ii) What is PEG-400?

- (iii) What are the advantages of PEG-400 over ionic liquids? 2+1+2

- (d) (i) How would you use the proton NMR to distinguish between 1-bromopropane and 2-bromopropane.

- (ii) How can you differentiate between the following pairs of molecules by application of IR spectroscopy?

(I) Methyl benzoate and o-hydroxy acetophenone

(II) o-hydroxy and p-hydroxy benzoic acids

2+(1.5+1.5)

- (e) Carry out the following conversions:

(i) Indole $\rightarrow$ Quinoline

(ii) Pyridine $\rightarrow$ 4-Bromo pyridine

2.5+2.5

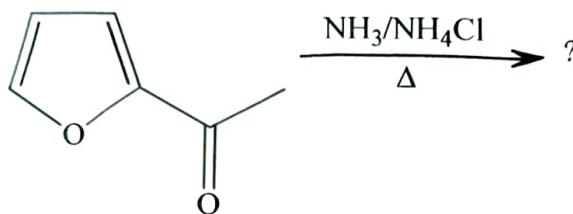
- (f) (i) Provide an explanation for the fact that under the same reaction conditions (EtONa in EtOH at $75^\circ C$), the *cis*-isomer of 4-tert-butylcyclohexyl tosylate undergoes E2 elimination, but the *trans* isomer does not.

- (ii) Pyrrole undergoes electrophilic substitution at 2-position whereas indole undergoes electrophilic substitution at 3-position. — Explain. 3+2

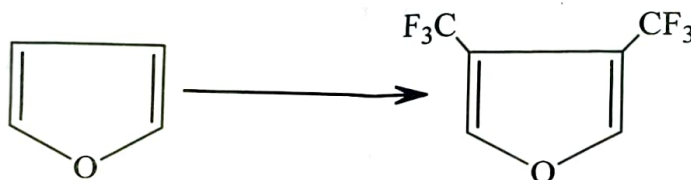
3. Answer any two questions from the following:

10×2=20

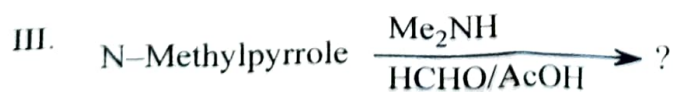
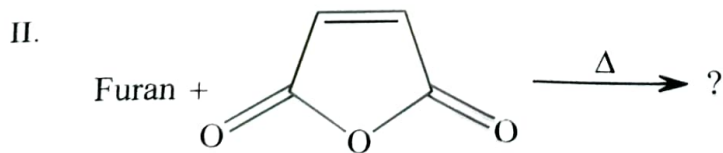
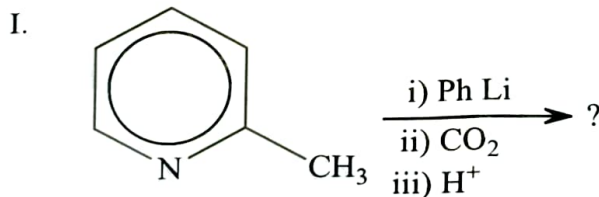
(a) (i) Write down the product of the following reaction with proper mechanism:



(ii) Convert:

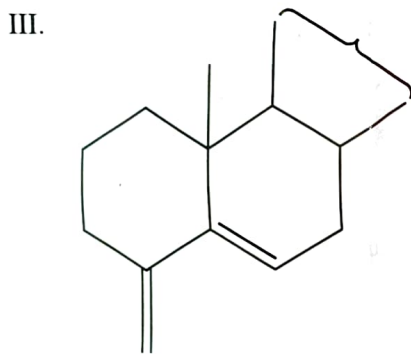
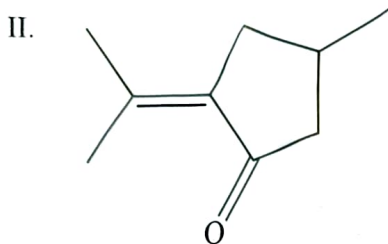
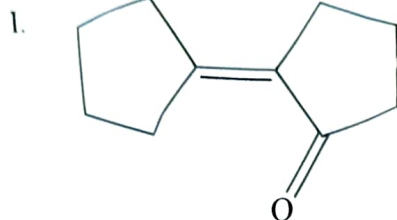


(iii) Write down the structure of the products formed by the following reactions:



2+2+(2+2+2)

(b) (i) Calculate the λ_{max} values of the following compounds:



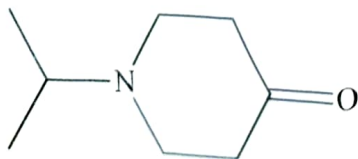
(ii) Using 60 MHz NMR spectrometer, a chemist observes the following proton signal.
Doublet, $J = 7 \text{ Hz}$ at $\delta_{\text{H}} 4.00$ (in ppm)

(I) What will be the frequency of the given signal in Hz unit?

(II) Where would this peak be located (in ppm and in Hz) in a 100 MHz spectrometer?

(III) What would be the coupling constant (J), in a 100 MHz spectrometer?

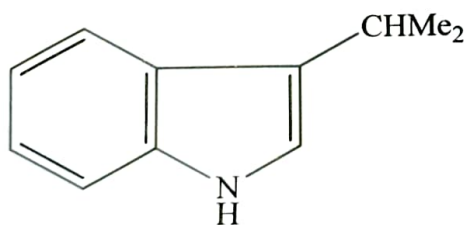
(iii) Assign the proton NMR spectrum of the following compound and explain the multiplicity of the signals.



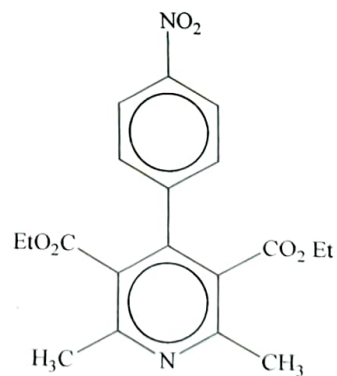
δ_{H} 1.08 (6H, d, J 7 Hz)
2.45 (4H, t, J 5 Hz)
2.80 (4H, t, J 5 Hz)
2.93 (1H, septet, J 7 Hz)

(iv) Why KBr is used as solid matrix for measurement of IR spectra of a solid sample?

- (c) (i) Is Wittig reaction example of 100% atom economic? Justify with suitable example.
(ii) Write down the green synthesis of adipic acid starting from cyclohexene as the starting precursor.
(iii) How can benzoin condensation be carried out without the use of cyanides?
(iv) Why ionic liquids are called designer solvents?
(v) Give two examples of organic reactions in ionic liquids. 2+2+2+2+2
- (d) (i) How would you differentiate between the following by spectroscopic method mentioned in the bracket.
(I) *o*-Xylene and *p*-Xylene (NMR)
(II) Benzaldehyde and Acetaldehyde (IR)
(ii) Arrange the following in order of increasing stretching frequency:
(I) C—H; N—H; O—H and S—H
(II) $\equiv$ C—H; —C—H and = C—H
(iii) Write down the product when pyrrole is treated with I₂ in aqueous KI.
(iv) What starting materials are required for the synthesis of Compound A by the Fischer Indole Synthesis and Compound B by Hantzsch synthesis, respectively?



(A)



(B)

(1+1)+(1+1)+2+(2+2)